How accurate LIGHT v0_4 is

The basic network of LIGHT v0_4, version 0.1.4, on 9 October 2026: v0_3 with the 5p rung of krypton, the 248-nm optics, KrF(B,C) + F, ion-ion recombination's limbs and bands, the ions' drift to the walls and the EM wave, with its energy ladder and its checks

Bottom line. LIGHT v0_4 reproduces the fitted KrF decays [1, 2] within 6.1 % rms (2.3 to 6.1 %) and the decays no fit used [4] within 11.7 % rms, as v0_2 and v0_3 do. At the Tahoma laser [5] it gives the small-signal gain to -7 % (5.91 against the measured 6.39 %/cm), the absorption, scored for the first time, to -26 % (0.82 against 1.11), the net gain to -4 %, the saturation intensity to +5 % and the formation efficiency to -4 %. No fit used Tahoma's numbers and no rate was set by a laser measurement.

Its own uncertainty, propagated from its sources (Section 1: 22 inputs, each at the two ends its source allows, combined in quadrature), is ± 39 % on the 248-nm share at the reference point and ± 33, ± 37 and ± 40 % on Tahoma's gain, absorption and net gain. The largest term everywhere but the absorption is the trimer's $(T_g/300)^n$, a band with no measurement behind it. Every Tahoma measurement lies within one quadrature unit of the model (the gain 0.3, the absorption 1.0, the net gain 0.1 units): at Tahoma the model's own uncertainty, not its error, is the limit.

The largest errors: Hakuta et al.'s C-A decays [6], 19.3 and 34.1 % rms; Burnham and Searles' KrF(B) lifetime [7], -22.3 %; the VUV yield of pure krypton at low pump [8], +32 % (v0_3's run, not repeated here); and Tahoma's absorption, -26 %. Kr_2F^* makes 73 % of that absorption at the cross section Hakuta et al. measured [9]; at the compilation value every KrF model carried, the absorption would be $\times 3.43$ the measurement. What v0_4 changes at the reference point is the 5s $\leftrightarrow$ 5p electron cycle on Kannari's fit [10, 11]: it turns 33.8 % of the input into heat and takes the 248-nm share from 4.67 to 5.8 % of E_{dep} (5.68 % under the kinetic EEDF). That fit has no direct measurement behind it, and its band runs to 0.

LIGHT v0_4 is v0_3 with what the audit of 8 October 2026 found missing against the measured data and the modern kinetic codes: the 5p rung of krypton (Kr^{**}, fed by the higher excitations and by $Kr_2^+ + e^-$, relaxed by Kr at a measured rate, radiating at the measured manifold lifetime, harpooning F_2 at a measured rate, pumped from Kr^* and returned by the electrons on Kannari's fit, its two stepwise ionizations bands at 0), KrF(B,C) + F as a band, the 248-nm optics from Kannari's absorber block with Hakuta et al.'s measured Kr_2F^* cross section, ion-ion recombination on Flannery's turnover with the Langevin limb of the measured mobilities and its yield and B/C bands, the ions' drift to the walls of a gap, and an EM wave absorbed by the Lorentz gas on the atoms plus Turnbull et al.'s electron-ion inverse bremsstrahlung with their model's error as a band. The kinetic EEDF carries the 5s $\leftrightarrow$ 5p collisions, the ions' Coulomb collisions and the wave's high-frequency field term. The 10 kinetic datasets have no electrons and no Kr^{**}, so v0_4 scores them exactly as v0_2, and `make_overview_v0_4.py` refuses to write a number that differs. Table 1 lists every test.

1 Overall accuracy and its uncertainty

Against the data. Table 1 scores every test; Table 2 reads the same numbers as errors, the model's deviation from the measurement in percent of the measurement (rms over a set of points), and names the largest.

test	measured	v0_4	v0_3	role of the data
energy ledger closure		below 2.0×10^{-14} with walls, 4.0×10^{-14} with the wave	below 10^{-13}	bookkeeping
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	the same	train
KrF(B,C) slow decay, Gadomski 1992 Figs. 1 and 2		rms 6.1 and 4.2 %	the same	train
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	the same	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	the same	held out
decay constants Xu 1993 printed, 50 and 100 Torr Kr	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	the same	structure check
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	the same	held out, in conflict
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	the same	held out, in conflict
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-8.4 %	the same	held out
Tahoma small-signal gain g_0, kinetic EEDF	6.39 %/cm	5.91, $\times0.93$	5.64, $\times0.88$	device, score only
the same, v0_2's closure		5.86, $\times0.92$	5.68, $\times0.89$	device, score only
Tahoma absorption α_0, kinetic EEDF	1.11 %/cm	0.82, $\times0.74$	not computed	device, score only
the same, v0_2's closure		0.82, $\times0.74$	0.84, $\times0.75$ (v0_4's cross sections on v0_3's densities)	device, score only
Tahoma net gain $g_0 - \alpha_0$, kinetic EEDF	5.28 %/cm	5.09, $\times0.96$		device, score only
Tahoma loss in pure Kr, 257.3 nm	0.67 %/cm	0.24, $\times0.36$ (248-nm cross sections)		rough check
Tahoma saturation intensity I_{sat}, v0_2's closure	1.4 ± 0.1 MW/cm^2	1.47, $\times1.05$	1.56, $\times1.11$ (kinetic)	device, score only
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$, v0_2's closure	0.27	0.260, $\times0.96$	0.266, $\times0.99$ (kinetic)	device, score only

Table 1: LIGHT v0_4 against the data, beside v0_3, every band at its default unless a row says otherwise. A percentage is the deviation from the data, signed for a single value and rms for a set. A $\times$ factor is the model over the measurement. "The same" is checked: the generator stops if v0_4's number differs from v0_2's. The absorption rows are new: Tahoma's Table 3-3 (beam stop in, 330 kW cm^{-3}) prints g_0, α_0 and $g_0 - \alpha_0$, and Table 3-2 the pure-Kr loss at 257.3 nm.

quantity	measured	v0_4	error
KrF(B,C) decays the fit used, Eden 1978 and Gadomski 1992 [1, 2]	four sets		within 6.1 % rms (2.3 to 6.1)
KrF(B,C) decays no fit used, Xu 1993 [4]	three figures		within 11.7 % rms
C/B emission ratio at 300 K [3]	0.16 ± 0.02		-0.5 %
its 230 K/300 K ratio [3]	1.30 ± 0.06		-8.4 %
decay constants, 50 and 100 Torr Kr [4]	5.0×10^7, 6.2×10^7 s^{-1}	$\times0.91$, $\times0.97$	-9, -3 %
Tahoma small-signal gain g_0 [5]	6.39 %/cm	5.91	-7 %
Tahoma absorption α_0	1.11 %/cm	0.82	-26 %
Tahoma net gain $g_0 - \alpha_0$	5.28 %/cm	5.09	-4 %
Tahoma saturation intensity	1.4 ± 0.1 MW/cm^2	1.47	+5 %
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$	0.27	0.260	-4 %
the largest errors			
C-A decays, Hakuta 1990 [6] (in conflict with the fitted sets)	two figures		19.3 and 34.1 % rms
KrF(B) lifetime, Burnham and Searles 1977 [7] (in conflict)	9.0 ± 0.6 ns		-22.3 %
VUV yield of pure Kr at low pump, Morozov 2008 [8], v0_3's run	0.42 ± 0.05	0.55	+32 %
Tahoma absorption α_0	1.11 %/cm	0.82	-26 %

Table 2: The scores of Table 1 as errors. A single value's error is signed, model less measurement over the measurement; a set's is the rms over its points. Tahoma's are under the kinetic EEDF; the rest have no electrons.

Propagated from the sources. The model's own uncertainty is what its inputs' uncertainties do to its outputs. Each input is set to the two ends its source allows and the outputs are recomputed (Table 3, under v0_2's closure; the kinetic one moves these outputs by about 1 %): each of the 17 bands to its range (a published spread, or 0 to an estimate where nothing is measured, Table 6); the 3 constants the model uses at a measured value whose source quotes an error and that no band covers, $Kr_2^+ + e^-$ (±10 % [15]), $Kr^+ + 2Kr$ (±9 % [24]) and the Kr_2F^* lifetime (±9 % [4]), to $1\pm$ that error; and the 2 cross sections with a quoted error analytically, σ_{se} ±12 % [23, 10] on g_0 and $\sigma(Kr_2F^*)$ ±20 % [9] on its share of α_0, which is ±15 % of α_0. Half the difference between an input's two ends is its term. The quadrature sum treats the terms as independent and each as one standard deviation, which a band's range is not, so it is an indicative number. The linear sums of the downward and of the upward excursions, every input pushing the same way, overestimate the spread (the inputs draw on the same KrF(B) and a share cannot fall below 0) and are printed as the ceiling on it, not as a bound. Not propagated: the electron-impact cross sections, which carry no quoted error, and the fitted KrF(B,C) set (the lifetimes, the B-C mixing, the quenches by e^-, F_2 and Kr, the trimer rate at 300 K), which the training decays pin at 6.1 % rms and Table 2 scores.

On the 248-nm share at the reference point (5.79 % of E_{dep}) the quadrature sum is ±2.28 points (±39 %), the linear sum ±5.17 points and the excursions sum to -7.96 and +2.39 points; the largest terms are the trimer's $(T_g/300)^n$ (±1.75), $Kr_2^* + F \rightarrow KrF(B) + Kr$ (±0.90), $KrF(B,C) + F \rightarrow Kr + 2F$ (±0.90), $e^- + Kr^* \rightarrow e^- + Kr^{**}$, pre-exponential ($\pm0.56$). On Tahoma's gain (5.86 %/cm) it is ±33 % in quadrature (-7.25 and +1.51 %/cm summed), the largest terms the trimer's $(T_g/300)^n$ (±1.52), KrF^* per ion-ion recombination (±0.73), σ_{se}, 248-nm stimulated emission (±0.70), $KrF(B,C) + F \rightarrow Kr + 2F$ (±0.46); the measurement is -8 % away, 0.3 quadrature units, within one unit. On the absorption (0.82 %/cm) it is ±37 % (-1.06 and +0.51 summed), the largest terms $Kr_2F^* + F_2$ (±0.18), $Kr_2F^* + F \rightarrow 2Kr + 2F$ (±0.16), $\sigma(Kr_2F^*)$ at 248 nm (±0.12), the trimer's $(T_g/300)^n$ (±0.10); the measurement is -26 % away, 1.0 units, within one unit. On the net gain (5.04 %/cm) it is ±40 % (-7.49 and +2.30 summed), the measurement -5 % away, 0.1 units. So at Tahoma the model's own uncertainty is larger than its errors: a third to two fifths of each output, against errors of a few percent on the gain and the net gain and -26 % on the absorption, and one unmeasured law, the trimer's $(T_g/300)^n$, carries most of it on the gain while the Kr_2F^* quenches and its cross section carry the absorption's. Measuring that

law, or the trimer rate at the operating temperature, would tighten the gain's uncertainty more than any other measurement.

input	its two ends	248 nm (% of E_{dep})	g_0 (%/cm)	α_0 (%/cm)	$g_0 - \alpha_0$
$Kr_2^* + F \rightarrow KrF(B) + Kr$	0 to 1.0×10^{-9} $cm^3\,s^{-1}$	5.79 to 7.59	5.86 to 6.21	0.82 to 0.82	5.04 to 5.39
$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0 to 5.0×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.80	5.86 to 5.86	0.82 to 0.82	5.04 to 5.04
$Kr_2F^* + F \rightarrow 2Kr + 2F$	0 to 1.8×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.93	5.86 to 6.00	0.82 to 0.50	5.04 to 5.50
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$, 300 K	1.7×10^{-32} to 5.5×10^{-32} $cm^6\,s^{-1}$	5.79 to 5.31	5.86 to 5.50	0.82 to 0.82	5.04 to 4.68
$Kr_2^* + F_2$, both channels	2.1×10^{-10} to 5.0×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.81	5.86 to 5.91	0.82 to 0.83	5.04 to 5.08
$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	1.5×10^{-10} to 5.0×10^{-10} $cm^3\,s^{-1}$	6.01 to 5.79	5.88 to 5.86	0.82 to 0.82	5.06 to 5.04
$Kr_2F^* + F_2$	1.2×10^{-10} to 4.3×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.96	5.86 to 6.05	0.82 to 0.46	5.04 to 5.59
the trimer's $(T_g/300)^n$	-3.32 to -0.5	5.79 to 2.29	5.86 to 2.81	0.82 to 1.03	5.04 to 1.78
KrF(B,C) + F → Kr + 2F	0 to 7.8×10^{-10} $cm^3\,s^{-1}$	5.79 to 3.99	5.86 to 4.93	0.82 to 0.75	5.04 to 4.18
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0×10^{-11} to 1.7×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.79	5.83 to 5.86	0.93 to 0.82	4.89 to 5.04
$e^- + Kr^* \rightarrow e^- + Kr^{**}$, pre-exponential	0 to 9.0×10^{-7} $cm^3\,s^{-1}$	4.67 to 5.79	5.69 to 5.86	0.84 to 0.82	4.84 to 5.04
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$, pre-exponential	0 to 1.5×10^{-7} $cm^3\,s^{-1}$	5.79 to 5.13	5.86 to 5.89	0.82 to 0.82	5.04 to 5.07
$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$, pre-exponential	0 to 2.0×10^{-7} $cm^3\,s^{-1}$	5.79 to 5.56	5.86 to 5.87	0.82 to 0.82	5.04 to 5.05
$Kr^{**} + F_2$, all channels	0 to 1.3×10^{-9} $cm^3\,s^{-1}$	5.76 to 5.79	5.85 to 5.86	0.82 to 0.82	5.03 to 5.04
KrF^* per ion-ion recombination	0.5 to 1	5.68 to 5.79	4.40 to 5.86	0.66 to 0.82	3.74 to 5.04
share of the ion-ion KrF^* in C	0 to 0.5	5.79 to 5.76	5.86 to 5.34	0.82 to 0.80	5.04 to 4.53
the wave's electron-ion absorption, a factor	0.9 to 1.1	5.79 to 5.79	5.86 to 5.86	0.82 to 0.82	5.04 to 5.04
$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	±10 % (Shiu and Biondi 1977, ±10 %)	5.78 to 5.79	5.87 to 5.85	0.82 to 0.82	5.05 to 5.04
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	±9 % (Grössl et al. 1981, $(2.2 \pm 0.2)\times10^{-31}$)	5.78 to 5.79	5.87 to 5.85	0.82 to 0.82	5.05 to 5.03
Kr_2F^* radiative lifetime	±9 % (Xu et al. 1993, 220 ± 20 ns)	5.79 to 5.78	5.86 to 5.86	0.84 to 0.80	5.02 to 5.06
σ_{se}, 248-nm stimulated emission	±12 % (Rokni et al. 1978, $(2.5 \pm 0.3)\times10^{-16}$)	–	5.16 to 6.56	–	4.34 to 5.74
$\sigma(Kr_2F^*)$ at 248 nm	±20 % (Hakuta et al. 1987, $(1.0 \pm 0.2)\times10^{-18}$)	–	–	0.70 to 0.94	4.92 to 5.16

Table 3: The propagated uncertainty's inputs, each alone at its two ends, the others at their defaults, and the four outputs at those ends (low end to high end). The reference point is the notes' (1 atm, 0.5 % F_2, 300 kW cm^{-3} for 1 μs); Tahoma is the 200 to 400-ns window under the beam drive. The cross sections enter analytically, so they have no reference-point column. 6 minutes of runs.

2 What v0_4 changes

The 5p rung. One lumped Kr** for the $4p^5 5p$ manifold at 11.30 eV, the Biagi group's own threshold. C1a to C1c make it, $Kr_2^+ + e^-$ makes it (Shiu and Biondi [15] saw the 5p lines), Kr relaxes it to Kr* at 1.7×10^{-10} $cm^3 s^{-1}$ (Gazeli et al. [12], the total loss of one 5p′ level in Kr, the band down to the Ar analogue 3.0×10^{-11} [25]), it radiates to Kr* at the measured manifold lifetime of 29.0 ns (the statistical mean of the ten measured level lifetimes [13]), it harpoons F_2 at 1.3×10^{-9} $cm^3 s^{-1}$ (Nelson and Setser [14], read from a search snippet of their abstract) with Kr*'s yield and B/C split, and the electrons pump it from Kr* at Kannari's 9.0×10^{-7} $T_e^{0.50}$ $\exp(-1.51/T_e)$ and return it by detailed balance. The two stepwise ionizations have no measurement (Vriens [16] is a calculation) and are bands at 0, up to 1.5×10^{-7} and 2.0×10^{-7} on Kannari's forms; `-ladder kannari` sets every rung band to Kannari's value.

The cycle in the kinetic equation. Kannari's $A\,T_e^{1/2}\exp(-C/T_e)$ is exactly the Maxwellian rate of $\sigma(\varepsilon) = \sigma_0(1 - C/\varepsilon)$ above C, with $\sigma_0 = 1.34\times10^{-14}$ cm^2, so the pair is a population process on Kr* with that pseudo cross section (threshold the ledger's gap, 1.385 eV) and its Klein-Rosseland reverse on Kr**. On 2400 cells to 40 eV it gives the printed law to 2.2×10^{-4} and the detailed-balance ratio to 1.2×10^{-5}. With the pair as LIGHT-side rates alone the kinetic energy balance missed LIGHT's by 0.7 of the input; in a 60-ns pulse with it inside, the field carries 1.003 to 1.007 of the input and the balances agree to 2.4×10^{-7}.

KrF(B,C) + F (QF, QFC): a band, 0 to Kannari's 'analogous to F_2' [10] 7.8×10^{-10} $cm^3 s^{-1}$. No measurement of it, or of any RgX* + X analogue, exists.

The optics. $g_0 = \sigma_B n_B$ with $\sigma_B = 2.6\times10^{-16}$ cm^2 and $\alpha = \sum_i \sigma_i n_i$ over 6 absorbers: F^- 5.6×10^{-18} (measured), Kr_2^+ 1.6×10^{-18} (an estimate), Kr** 6.0×10^{-18} (a calculation), F_2 1.2×10^{-20}, Kr* 1.0×10^{-19} (compiled) and Kr_2F^* 1.0×10^{-18} cm^2, Hakuta et al.'s direct measurement [9] in place of the compilation's 6.0×10^{-18} [10, 22]; the cross sections are Kannari's Table I, F^- from Mandl [20], Kr** from Hyman [21]. A diagnostic: LIGHT has no photon field, so the absorbers change no density.

Ion-ion recombination. Flannery's turnover (as Kannari [10] writes it) as the harmonic mean of REBUILD's three-body limb, 4.82×10^{-6} N at 1 amagat, and the Langevin limb of the measured reduced

mobilities (Ellis et al. [17, 18]; Kr^+ 0.84, Kr_2^+ 1.22, F^- 2.29 $cm^2/V\,s$), $5.66\times10^{-6}/N$ for $Kr^+ + F^-$ and $6.35\times10^{-6}/N$ for $Kr_2^+ + F^-$, where REBUILD's was $8.2\times10^{-6}/N$ for both. The 1-amagat rate moves from 3.04×10^{-6} to 2.61×10^{-6} (Kr^+) and 2.74×10^{-6} (Kr_2^+) cm^3s^{-1}. The KrF* yield (ii_yield, 1 by default, down to 0.5) and its C share (ii_f_c, 0 as in v0_3, up to 0.5) are bands on the three product rows.

The walls. In a gap of 1 cm the cations leave at the ambipolar rate of the electropositive global model [33], D_a/Λ^2, $D_a = \mu_i k(T_e + T_g)/e$ with the mobilities above and Λ the gap over π, F^- at its free diffusion; each lost pair takes its formation energy and 2 T_e to the wall and 1 T_e to the gas through the ambipolar field. At the reference point's mid-pulse the rate is 63 s^{-1}, the walls take 9.4×10^{-6} % of the input and the 248-nm share is 5.79 %; in a 0.1-cm gap at 0.1 atm they take 0.069 %. No 1-D number checks this closure yet: 1Dv0_2's walls line at the same point is 0.036 % on 256 uniform cells, but that line also holds its transport step's mean-energy cut in the wall cells, and 1Dv0_3, whose sheaths are resolved, prints no walls line of its own.

The wave. A second input whose absorbed power is an output. The electron-atom part is v0's Lorentz-gas kernel on the Biagi and SIGLO momentum transfer [32], the electron-ion part Turnbull et al.'s inverse bremsstrahlung [28, 29] (the Sommerfeld logarithm with the Gaunt-factor fit they take from Pradler and Semmelrock, ion screening at the ion-sphere radius, the Langdon factor [30]), with their stated 10 % error as the band ib_scale (0.9 to 1.1) on the electron-ion share of the power. Under the kinetic closure the wave is Hagelaar and Pitchford's [31] high-frequency field term with Turnbull's per-energy cross section in its momentum transfer, and for a Maxwellian handed to the solver that term gives the kernel's atom part to 6.7×10^{-3} and its ion part to 1.0×10^{-3} (at T_e = 1.5 eV, n_e of 10^{15} cm^{-3}). The ion part here is the marginal one, the power with the ions in the momentum transfer less the power without them, and that is what the band acts on under both closures. It is not the kernel's printed electron-ion share, ν_{ei}'s share of the numerator: at 1 atm and 95 GHz ν_{en} is 4.4 times ω, the wave is collisional, and the printed share is 1.49 times the marginal part (the two agree only for $\nu_{en} \ll \omega$). Two double-counting rules: the Langdon factor is applied under the Maxwellian closures only, and the ions enter each field term once (the DC stand-in field sees them Debye-screened, the wave with the IB logarithm). At the reference point a wave of 1500 V/cm rms at 95 GHz deposits 0.024 $J\,cm^{-3}$ beside the 0.3 given (7.5 % of all input), the band moving it by 0.05 %; the ions' marginal share at mid-pulse (T_e 5.46 eV, n_e 1.13×10^{14} cm^{-3}, ν_{en}/ω 9.6) is 0.03 % (the printed share 0.06 %) with a Langdon factor of 0.9816, and at 10^{15} and 10^{16} cm^{-3} and 1.5 eV it is 1.46 and 7.74 % at 95 GHz, 0.33 and 2.57 % at 1 THz. At 1 THz a field of 3 kV/cm absorbs 1.7×10^{-5} $J\,cm^{-3}$ in the microsecond. `ib.py` is held to the IB project's own implementations by 219 checks.

3 How the numbers were made

Every number in this note is written by `docs/make_overview_v0_4.py` from v0_4 and from `ML/`, or by `docs/make_system_map_v0_4.py` (the reference point under v0_2's closure, its bands and its optics, which the system map draws), and a test fails when those files no longer match the model. ML runs v0_4 when `LIGHT_VERSION` names it. Tahoma runs under the prototype beam drive, a test fixture in `docs/` and not part of the model, in ML's configuration (v0_2's: the default network with every band at its default and n_e free from a seed of 1.0×10^{10} cm^{-3}). The kinetic EEDF takes minutes a run, so Tahoma's variants, its saturation intensity and its formation efficiency run on v0_2's closure here (the kinetic closure moves v0_4's Tahoma numbers by about 1 %, Table 1), and the kinetic numbers are the reference point and Tahoma's default, 10 and 12 minutes each.

4 Tahoma and the optics

v0_2's closure	g_0 (%/cm)	α_0 (%/cm)	$g_0 - \alpha_0$	T_e (eV)	n_e (cm^{-3})
measured [5]	6.39	1.11	5.28		
every band at its default	5.86 ($\times$0.92)	0.82 ($\times$0.74)	5.04 ($\times$0.95)	1.18	4.51×10^{14}
the rung's bands at Kannari's values (`-ladder kannari`)	5.87 ($\times$0.92)	0.84 ($\times$0.76)	5.03 ($\times$0.95)	1.17	4.61×10^{14}
the 5s $\leftrightarrow$ 5p cycle off (`k_5s5p` = 0)	5.69 ($\times$0.89)	0.84 ($\times$0.76)	4.84 ($\times$0.92)	1.60	6.32×10^{14}
Kr** + Kr at the band's low end	5.83 ($\times$0.91)	0.93 ($\times$0.84)	4.89 ($\times$0.93)	1.26	4.82×10^{14}
KrF(B,C) + F at Kannari's analogue	4.93 ($\times$0.77)	0.75 ($\times$0.68)	4.18 ($\times$0.79)	1.18	4.5×10^{14}
Kr_2^* + F at the top of its band	6.21 ($\times$0.97)	0.82 ($\times$0.74)	5.39 ($\times$1.02)	1.19	4.49×10^{14}
the trimer on $T_g^{-1/2}$	2.81 ($\times$0.44)	1.03 ($\times$0.93)	1.78 ($\times$0.34)	1.19	4.48×10^{14}
ion-ion KrF* yield at 0.5	4.40 ($\times$0.69)	0.66 ($\times$0.59)	3.74 ($\times$0.71)	1.19	4.49×10^{14}
ion-ion C share at 0.5	5.34 ($\times$0.84)	0.80 ($\times$0.72)	4.53 ($\times$0.86)	1.18	4.52×10^{14}
v0_3	5.68 ($\times$0.89)	0.84 ($\times$0.75)	4.85 ($\times$0.92)	1.60	6.33×10^{14}
v0_4, kinetic EEDF	5.91 ($\times$0.93)	0.82 ($\times$0.74)	5.09 ($\times$0.96)	1.16	4.43×10^{14}

Table 4: Tahoma over the 200 to 400 ns window: v0_4 with every band at its default and each variant alone, v0_3, and v0_4 under the kinetic EEDF. The absorption by species with the defaults: Kr_2F^* 0.60, F^- 0.09, F_2 0.06, Kr** 0.03, Kr_2^+ 0.03, Kr* 0.02 %/cm (v0_2's closure), Kr_2F^* 0.60, F^- 0.09, F_2 0.06, Kr** 0.03, Kr_2^+ 0.03, Kr* 0.02 (kinetic); pure Kr, no F_2: Kr* 0.11, Kr** 0.10, Kr_2^+ 0.04.

The absorption is the first LIGHT has scored. With Kannari's compilation value for Kr_2F^* it came out 3.81 %/cm, $\times$3.43 the measurement, which sent the audit to the literature: Hakuta, Komori, Mukai and Takuma [9] measured 1.0×10^{-18} cm^2 at 248 nm by fluorescence suppression and wrote that it is 'about 1/5 of the value assumed in the current kinetic model'. With the measurement α_0 is $\times$0.74 of Tahoma's, Kr_2F^* 73 % of it, and the net gain $\times$0.95. The pure-Kr loss Tahoma measured at 257.3 nm, 0.67 %/cm, LIGHT gives as 0.24 with 248-nm cross sections, a rough check (Kr_2^+'s 1.6×10^{-18} cm^2 is itself an estimate). Under the beam the 5s $\leftrightarrow$ 5p cycle cools the electrons (1.18 against v0_3's 1.60 eV) and lowers n_e, and the gain rises by a few percent (Table 4's factors are the model over the measurement: $\times$0.92 with the cycle, $\times$0.89 without). The rung's own bands move it by under 1 % ($\times$0.92 at Kannari's values, $\times$0.91 with Kr** + Kr at its low end); the bands that move it are v0_2's and the new quenches, the trimer's temperature law ($\times$0.44), the ion-ion yield ($\times$0.69), KrF(B,C) + F at Kannari's analogue ($\times$0.77), the ion-ion C share ($\times$0.84) and Kr_2^* + F ($\times$0.97), none of them with a measurement behind the band. In the window under the kinetic EEDF LIGHT's mean energy is the kinetic one at every update, held by a field of at most 0.11 Td, and the two energy balances agree to 4.6×10^{-4} of the beam's electron power. ML's formation run gives g_0 5.86 %/cm with its own channels, I_{sat} 1.47 MW/cm^2 and $g_0 I_{sat}/P_{in}$ = 0.260.

5 The reference point

At the notes' reference point, 1 atm, 0.5 % F_2, 300 kW cm^{-3} for 1 μs and run to 2 μs, v0_4 gives these shares of the deposited energy:

	v0_4, kinetic	v0_4, v0_2's closure	v0_3, kinetic	v0_3, v0_2's closure
248 nm, KrF B-X	5.68	5.8	4.79	4.67
272 nm, KrF B-A	0.181	0.2	0.153	0.149
275 nm, KrF C-A	0.825	0.8	0.735	0.712
420 nm, Kr_2F^*	4.28	3.8	4.96	4.87
147 nm, Kr_2^*	5.3	4.5	19.7	20.0
near IR, 5p $\rightarrow$ 5s	0.27	0.3	–	–
all bands	16.5	15.5	30.3	30.4
heat, reactions	71.9	71.6		
to Kr^*, gross	65.4	60.6	99.4	98.3
to Kr^+	3.9	3.2	26.0	26.8
KrF(B) formed	24.3	23.0	24.6	23.8
elastic heating	2.0	3.4	5.3	6.2
g_0, 500 to 1000 ns (%/cm)	3.73	4.12		
α (%/cm)	0.68	0.63		
T_g at the end (K)	735	742	650	649
T_e, pulse mean (eV)	2.97	5.81		

Table 5: The reference point, percent of E_{dep} unless a row says otherwise. v0_4's v0_2-closure column is the system map's run.

What moves the result is the 5s $\leftrightarrow$ 5p cycle. On Kannari's fit the electrons pump 81.0 % of the Kr^* formed up to the rung (78.4 % under the kinetic EEDF), the rung relaxes 96.1 % of it to Kr^* on Kr and the 1.385 eV of every cycle heats the gas: the cycle takes 33.8 % of the input, cuts the Kr^* population and with it the Kr_2^* route, and the 248-nm share rises as the harpoon and the trimer share what is left. The near-IR 5p $\rightarrow$ 5s band is 0.3 %: the measured relaxation outruns the 29.0-ns radiation. With the rung neutralised (the cycle off, the rung relaxed at once) the share is 4.67 %, v0_3's; with every rung band at Kannari's value 5.18 %. Under the kinetic EEDF the field carries 1.001 to 1.004 of the input through the pulse's second half and the two energy balances agree to 8.0×10^{-6}; the run takes 414 kinetic solutions, 277 in the pulse, with E/N between 1.56 and 45 Td, and closes its ledger below 10^{-14}.

Every band's two ends under v0_2's closure, one band at a time (`run_v0_4.py -bands`):

band	default	range	248 nm	147 nm	g_0	α	T_g (K)
the trimer's $(T_g/300)^n$	-3.32	-3.32 to -0.50	5.79 to 2.29	4.5 to 4.3	4.12 to 1.26	0.63 to 0.96	742 to 754
KrF(B,C) + F $\rightarrow$ Kr + 2F	0	0 to 7.8×10^{-10} $cm^3\,s^{-1}$	5.79 to 3.99	4.5 to 4.4	4.12 to 2.42	0.63 to 0.48	742 to 761
Kr_2^* + F $\rightarrow$ KrF(B) + Kr	0	0 to 1.0×10^{-9} $cm^3\,s^{-1}$	5.79 to 7.59	4.5 to 0.2	4.12 to 5.88	0.63 to 0.64	742 to 757
e^- + Kr^* $\rightarrow$ e^- + Kr^{**}, pre-exponential	9.0×10^{-7}	0 to 9.0×10^{-7} $cm^3\,s^{-1}$	4.67 to 5.79	20.0 to 4.5	2.56 to 4.12	0.80 to 0.63	649 to 742
e^- + Kr^* $\rightarrow$ Kr^+ + $2e^-$, pre-exponential	0	0 to 1.5×10^{-7} $cm^3\,s^{-1}$	5.79 to 5.13	4.5 to 2.3	4.12 to 3.71	0.63 to 0.55	742 to 767
Kr^* + 2Kr $\rightarrow$ Kr_2^* + Kr, 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32} $cm^6\,s^{-1}$	5.79 to 5.31	4.5 to 10.6	4.12 to 3.80	0.63 to 0.68	742 to 706
e^- + Kr^{**} $\rightarrow$ Kr^+ + $2e^-$, pre-exponential	0	0 to 2.0×10^{-7} $cm^3\,s^{-1}$	5.79 to 5.56	4.5 to 3.9	4.12 to 3.96	0.63 to 0.60	742 to 749
Kr^* + Kr^* $\rightarrow$ Kr^+ + Kr + e^-	5.0×10^{-10}	1.5×10^{-10} to 5.0×10^{-10} $cm^3\,s^{-1}$	6.01 to 5.79	5.6 to 4.5	4.34 to 4.12	0.65 to 0.63	732 to 742
Kr_2F^* + F_2	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.96	4.5 to 4.5	4.12 to 4.22	0.63 to 0.39	742 to 754
Kr_2F^* + F $\rightarrow$ 2Kr + 2F	0	0 to 1.8×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.93	4.5 to 4.5	4.12 to 4.24	0.63 to 0.26	742 to 757
KrF^* per ion-ion recombination	1	0.5 to 1	5.68 to 5.79	4.5 to 4.5	4.04 to 4.12	0.62 to 0.63	743 to 742
share of the ion-ion KrF^* in C	0	0 to 0.5	5.79 to 5.76	4.5 to 4.5	4.12 to 4.11	0.63 to 0.63	742 to 742
Kr^{**} + Kr $\rightarrow$ Kr^* + Kr	1.7×10^{-10}	3.0×10^{-11} to 1.7×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.79	5.5 to 4.5	4.09 to 4.12	1.05 to 0.63	727 to 742
Kr^{**} + F_2, all channels	1.3×10^{-9}	0 to 1.3×10^{-9} $cm^3\,s^{-1}$	5.76 to 5.79	4.4 to 4.5	4.11 to 4.12	0.62 to 0.63	743 to 742
Kr_2^* + F_2, both channels	2.1×10^{-10}	2.1×10^{-10} to 5.0×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.81	4.5 to 3.3	4.12 to 4.11	0.63 to 0.66	742 to 747
Kr_2^* + Kr_2^* $\rightarrow$ Kr_2^+ + 2Kr + e^-	0	0 to 5.0×10^{-10} $cm^3\,s^{-1}$	5.79 to 5.80	4.5 to 3.4	4.12 to 4.10	0.63 to 0.63	742 to 749
the wave's electron-ion absorption, a factor	1	0.9 to 1.1	5.79 to 5.79	4.5 to 4.5	4.12 to 4.12	0.63 to 0.63	742 to 742

Table 6: Each band alone at its two ends, the others at their defaults (17 bands, 7 with no measurement and a default of 0). The columns are the 248-nm and 147-nm shares of E_{dep} (%), the 500-to-1000-ns gain and absorption (%/cm) and the final gas temperature, low end to high end. With every band at its default the 248-nm share is 5.8 %; no band but the four named in the text moves it by more than 0.66 points.

6 The energy ladder

v0_4's ladder (Figure 1) adds the 5p rung, 11.3 eV above Kr and 1.39 eV above Kr^*, with K76 and K76r between the two, KPS and K27X as quenches, PRAD's 1.39-eV photon, the two stepwise ionizations

and the harpoon from 5p; $Kr_2^+ + e^-$ now lands on it, leaving 1.55 eV to the gas where v0_3 left 2.93. KrF(B) lies 5.54 eV below the ion pair $Kr^+ + F^-$ and 5.06 eV above Kr + F, and the 248-nm photon carries 5.00 eV of that.

7 Where v0_4 is not yet reliable

- The 5s $\leftrightarrow$ 5p rate is a fit, not a measurement, and it decides the partition. Jung et al. [26, 27] measured the cross sections out of the J = 2 metastable into the ten 5p levels; their absolute values are not among the sources held. Band k_5s5p runs to 0 (4.67 to 5.79 % at 248 nm).
- KPS is one level's total loss carried to the manifold, its product unresolved; K27 is a search snippet of an abstract. Both are flagged in the rows.
- The stepwise ionizations and the 5p harpoon are LIGHT-side rates, outside the Boltzmann equation (0 by default). $Kr_2^+ + e^-$ makes the rung entirely; Shiu and Biondi say 'mostly'.
- The absorbers are a diagnostic; Kr_2^+'s and Kr^{**}'s cross sections are an estimate and a calculation, and the pure-Kr loss is checked with 248-nm values against a 257-nm measurement.
- The wall loss is the electropositive global-model closure, stated and reported. The wave's field is the one inside the plasma: no propagation, reflection or skin depth.
- Everything in v0_3's list: the steady EEDF after the pulse, the first cell, the thermal electrons' grid bias, $Kr_2^* + F$, the trimer above its measured range, Hakuta's and Burnham and Searles' conflicts, and the pure-krypton VUV yield at low pump.

A Reaction rates

v0_4's rows are v0_2's (its overview's Appendix A) and these, each with the value the model ran at the reference point and its source, as `v0_4/network.py` cites it; `v0_4/SOURCES.md` section 7 says how each was read.

row	reaction	value	source
C1a–c	$e^- + Kr \rightarrow e^- + Kr^{**}$	Biagi groups [32]	the higher excitations end on the rung (v0_3: on Kr^*)
R11p	$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	$1.6\times10^{-6}\,(300\,\mathrm{K}/T_e)^{0.55}$	Shiu and Biondi [15]; the 5p lines in the afterglow
KPS	$Kr^{**} + Kr \rightarrow Kr^* + Kr$	1.7×10^{-10} cm^{3}s^{-1}	Gazeli et al. [12], $5p'[3/2]_2$ in Kr; band to 3.0×10^{-11} (Jancaitis [25], Ar analogue)
PRAD	$Kr^{**} \rightarrow Kr^* + h\nu$	τ = 29.0 ns	Dzierżęga et al. [13] Table 1, ten measured lifetimes
K76	$e^- + Kr^* \rightarrow e^- + Kr^{**}$	$9.0\times10^{-7}\,T_e^{0.50}\,e^{-1.51/T_e}$	Kannari [10] [76], Jacob and Mangano's fit [11]; band to 0
K76r	$e^- + Kr^{**} \rightarrow e^- + Kr^*$	detailed balance, g 12/36	the statistical weights
KSTEP_S	$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	0; band to 1.5×10^{-7}	Kannari [79], Vriens's calculation [16] (7.0×10^{-8}); REBUILD's template
KSTEP_P	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	0; band to 2.0×10^{-7}	Kannari [79], Vriens [16]
K27, K27C, K27X	$Kr^{**} + F_2 \rightarrow$ KrF(B), KrF(C), Kr + 2F	1.3×10^{-9} cm^{3}s^{-1}, split as Kr^*'s	Nelson and Setser [14] (snippet); Kannari's 8.1e-10 a mark
QF, QFC	KrF(B,C) + F $\rightarrow$ Kr + 2F	0; band to 7.8×10^{-10}	Kannari [10], 'analogous to F_2'
K26, K108	Kr^+, $Kr_2^+ + F^- \rightarrow$ KrF(B)	2.61×10^{-6}, 2.74×10^{-6} cm^{3}s^{-1} at 1 amagat	Flannery's turnover [10]: REBUILD's three-body limb, the Langevin limb of Ellis's mobilities [17, 18]
K26C, K108C, K26X, K108X	the C and dissociating shares	bands ii_f_c, ii_yield	Rokni, Jacob and Mangano [19] (relative); REBUILD's split
σ at 248 nm	F^-, Kr_2^+, Kr^{**}, F_2, Kr^*, Kr_2F^*	5.6×10^{-18}, 1.6×10^{-18}, 6.0×10^{-18}, 1.2×10^{-20}, 1.0×10^{-19}, 1.0×10^{-18} cm^2	Kannari's Table I [10] (its refs. 81, 82, 62, 84, 16: [20, 19, 21, 22]); Hakuta et al. [9] for Kr_2F^*
walls	D_a/Λ^2	K_0 0.84, 1.22, 2.29 cm^2/V s	Ellis et al. [17, 18]
wave	Lorentz gas + Turnbull IB	band ib_scale 0.9 to 1.1	Turnbull et al. [28, 29]; Hagelaar and Pitchford [31]

Table 7: The rows and constants v0_4 adds. Rates in cm^3/s; a 0 is a band whose default is 0, no measurement.

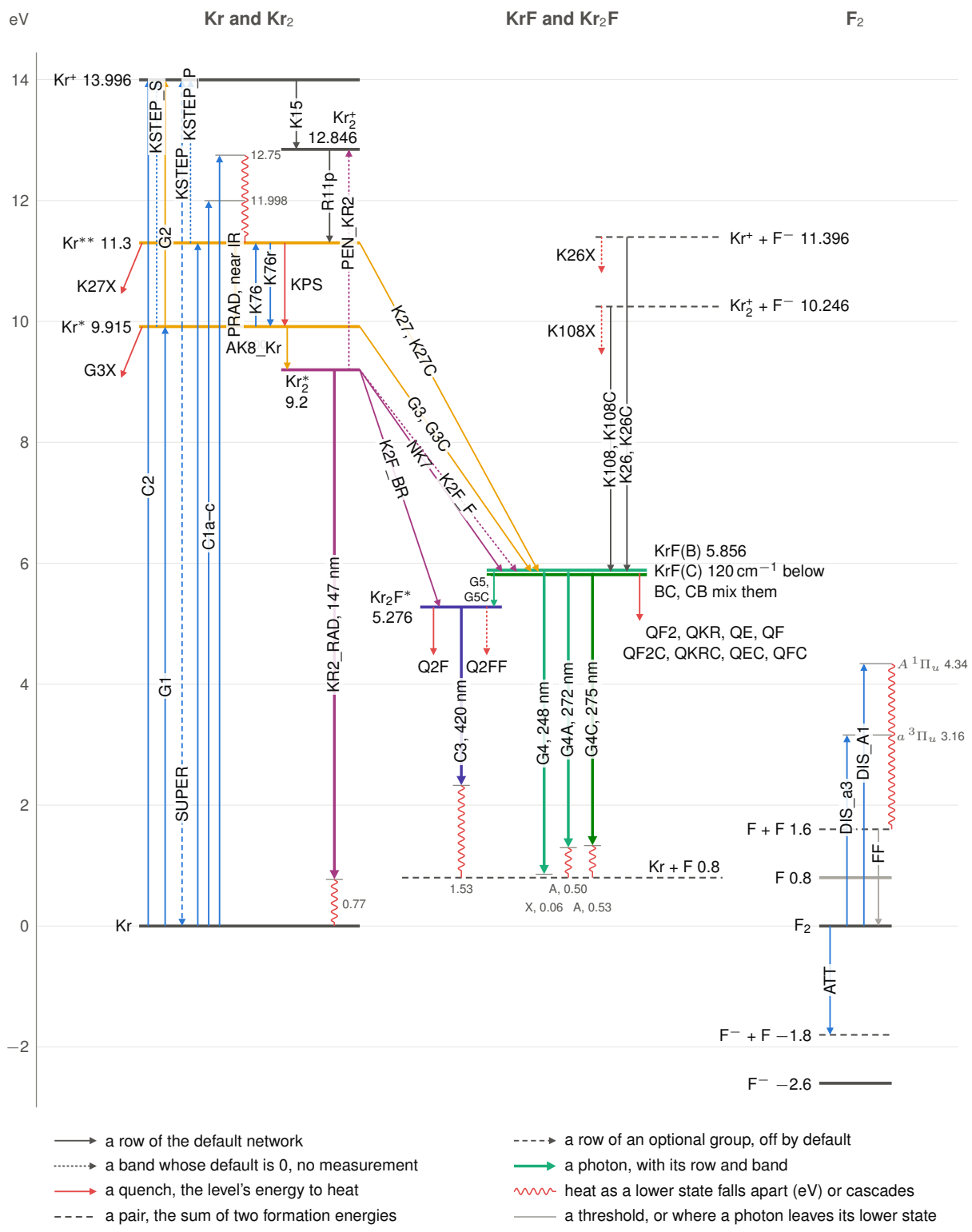


Figure 1: The energy ladder of LIGHT v0_4, version 0.1.4, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id. It is solid in the default network and dotted for a band whose default is 0, Q2FF, K26C, K26X, K108C, K108X, K2F_F, PEN_KR2, KSTEP_S, KSTEP_P, QF and QFC. The rows of the optional group `electron` are dashed. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2, K76r and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 120 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

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